



6.5.1 Systematic derivation of backward error analyses

Throughout the course, we have pointed out that the FLAME notation facilitates the systematic derivation of linear algebra algorithms. The papers

- [6] Paolo Bientinesi, Robert A. van de Geijn, Goal-Oriented and Modular Stability Analysis, SIAM Journal on Matrix Analysis and Applications, Volume 32 Issue 1, February 2011.
- [7] Paolo Bientinesi, Robert A. van de Geijn, The Science of Deriving Stability Analyses, FLAME Working Note #33. Aachen Institute for Computational Engineering Sciences, RWTH Aachen. TR AICES-2008-2. November 2008. (Technical report version of the SIAM paper, but with exercises.)

extend this to the systematic derivation of the backward error analysis of algorithms. Other publications and texts present error analyses on a case-by-case basis (much like we do in these materials) rather than as a systematic and comprehensive approach.